

03- MECHANICAL EQUIVALENT OF HEAT

PRINCIPLE

In this experiment, a metal test body is rotated and heated by the friction due to a tensed band of synthetic material. The mechanical equivalent of heat for problem 1 is determined from the defined mechanical work and from the thermal energy increase deduced from the increase of temperature. Assuming the equivalence of mechanical work and heat, the specific thermal capacity of aluminium and brass is determined.

TASKS

1. Determination of the mechanical equivalent of heat.
2. Determination of the specific thermal capacity of aluminium and brass.



Fig.1: Experimental set-up

THEORY

For a long time it was debated as to whether the heat of a system, which used to be defined in terms of its temperature, was a form of energy or an independent magnitude which is conserved. During the first half of the nineteenth century, it was proven that mechanical energy due to friction is completely converted to heat, independently of the course of the transformation process and of the physical and chemical characteristics of the material used. Heat was accordingly defined as the energy of disorganized, macroscopically invisible molecular movements.

The quotient between realized mechanical work W and the quantity of heat Q generated through friction is called the mechanical equivalent of heat. In this experiment, mechanical work is performed by rotating the friction cylinder against the synthetic friction band's sliding frictional force. As weight M (cf. Fig. 2) is not accelerated when the crank is rotated, weight $F_G = M_g = 49 \text{ N}$ (M : mass of the weight, g : terrestrial gravitational acceleration) on one side and the sliding frictional force acting together with force on the suspension of the dynamometer on the other side, must cancel each other:

$$F_R = F_G - F_D \quad (1)$$

Friction work W is thus determined from the friction path of (r : radius of the cylinder, n : number of turns) and is found to be:

$$W = 2\pi r n F_R = 2\pi r n F_G - F_D \quad (2)$$

Whilst the synthetic band slides over the cylinder, a large portion of mechanical work is converted to thermal energy ΔQ through inelastic deformations of the friction material. Accordingly, the temperature T of the friction cylinder increases by a temperature difference ΔT :

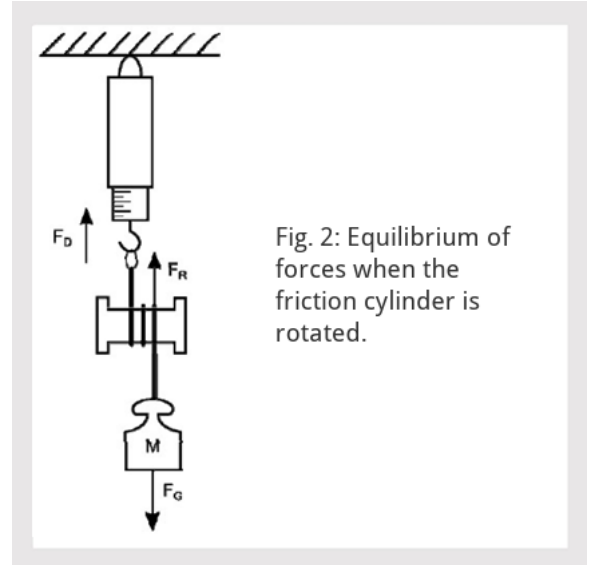
$$\Delta Q = C_{tot} \cdot \Delta T \quad (3)$$

C_{tot} is the total thermal capacity of the heated parts. It is the sum of the thermal capacity C_{cyl} of the friction cylinder (m : mass, specific thermal capacity (CuZn: $c = 0.385 \text{ J / gK}$)):

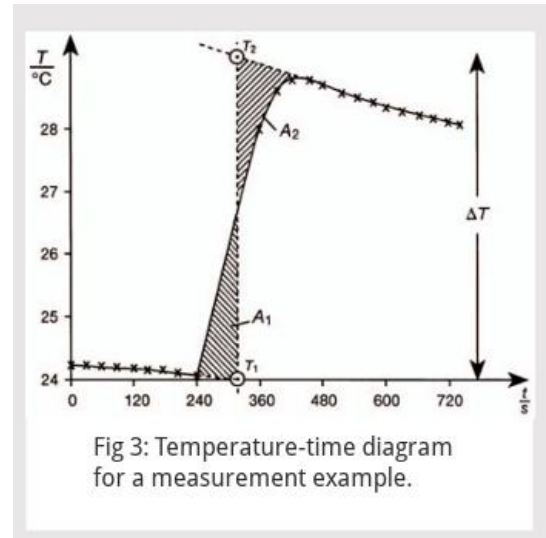
$$C_{cyl} = c \cdot m \quad (4)$$

Of the thermal capacity of the friction band ($C_{band} = 4 \text{ J/K}$) and of the thermal capacity of the thermometer ($C_{th} = 4 \text{ J/K}$). If the mass of the friction cylinder is $m = 640 \text{ g}$, total thermal capacity amounts to:

$$C_{tot} = C_{cyl} + C_{band} + C_{th} = 254 \frac{\text{J}}{\text{K}} \quad (5)$$



The thermal capacities of the insulated bearing and of the heat conducting paste were neglected. Fig. 3 displays a temperature-time diagram for an example of measurement. The shape of the graph makes it clear, that thermal energy is continuously given off to the environment while the crank is being turned. Hence, temperature difference ΔT is determined as is generally done for the determination of specific heat: momentary temperature compensation is extrapolated through prolongation of both straight graph branches. If a



a parallel line to the temperature axis is drawn in such a way that two equal surfaces A_1 and A_2 are formed, the difference of the ordinates of the points of intersection of the two straight lines is the temperature difference T sought after.

The cylinder is cooled mainly due to convection. In this case we can use the Newton's law of cooling as a good approximation: "The rate of heat loss of a body is proportional to the difference in temperatures between the body and its surroundings". In this approximation the temperature difference in Fig. 3 corresponds to the heat loss rate (first derivative of total heat amount). The integral of temperature difference thus equals to the total amount of lost heat, which graphically corresponds to the area confined under the temperature curve in Fig. 3. There are two curves in Fig. 3. The first one (real experiment) corresponds to the mechanical work slowly transferred into heat. The second curve (imaginary experiment) shows instantaneous transfer. In both experiments the cylinder has the same amount of heat in the beginning of the experiment and after heating due to friction. Mechanical work is the same for both curves. Thus the amount of lost heat (area under curves) due to convection must be the same for both curves. This is possible only if , $A_1 = A_2$ which uniquely defines the position of the vertical line (instantaneous process) in Fig. 3.

Exponential cooling process can be approximated with straight lines for short time intervals.

In the case of a measurement example where the residual load on the dynamometer is $F_D = 3$ N, work after 200 rotations of the friction cylinder amounts to:

$$\Delta W = 1301 \text{ J}$$

According to equation (3), the resulting temperature increase of the friction cylinder, $\Delta T = 5.1$ K, produces the following amount of thermal energy:

$$\Delta Q = 1296 \text{ J}$$

$$\frac{\Delta W}{\Delta Q} = 1 \quad (6)$$

If the mechanical heat equivalent is calculated, the quotient (1.003) amounts to 1 within the limits of measurement error. As heat is a form of energy, the total amount of mechanical energy must be transformed into an equal amount of heat energy according to the law of conservation of energy. In fact, the mechanical equivalent of heat must thus amount to:

Similar experiments, in which electric energy is converted to thermal energy, show that the corresponding electric equivalent of heat $\Delta W_{el}/Q$ also amounts to 1. These results are comprehensively summarized in the first law of thermodynamics.

The change of internal energy ΔU of a system is equal to the amount of heat energy ΔQ and to the amount of work ΔW supplied from outside, independently of the supply procedure:

$$\Delta U = \Delta Q + \Delta W \quad (7)$$

If, inversely, equation 6 is assumed ($\Delta Q = \Delta W$), the specific thermal capacity can be determined with this equipment. If equations 3 and 5 are transformed according to the thermal capacity of the friction cylinder, the following result is obtained:

$$C = -8 \frac{J}{gK} \quad (8)$$

The specific thermal capacity, defined as the quotient between the thermal capacity of a body C and its mass:

$$c = C/m \quad (9)$$

is a material constant, which at normal ambient room temperatures and atmospheric pressure depends very little on absolute temperature. It is to be noted from equation 9 that the thermal capacity of a homogeneous material is strictly proportional to its mass.

An example measurement with a rest load $F = 4 \text{ N}$ on the dynamometer, 200 turns of the crank, and a temperature increase of $\Delta T = 2.5 \text{ K}$ yielded a thermal capacity of

$$C = 501 \text{ J/gK}$$

for a brass cylinder of 1280 g. (that is actually about twice the thermal capacity of a cylinder of half the weight (247 J / gK)). The specific thermal capacity calculated with equation 9 is:

$$C_{brass} = 501 \text{ J/gK}$$

The measurement of the aluminium cylinder yielded the following value:

$$C_{al} = 0.870 \text{ J/gK}$$

as compared to the theoretical value $C_{al} = 0.902 \text{ J/gK}$ for the specific thermal capacity of aluminium at room temperature.

Equipment

Friction Cylinder CuZn (Brass), $m=1.28 \text{ kg}$, Friction Cylinder Aluminum, $m=0.39 \text{ kg}$, Digital stopwatch, Weight 5 kg , Dynamometer, Thermocouple, Multimeter, Thermometer, Heat transfer paste.

Set-up and Prodecure:

Part I

1. First, secure the friction cylinder and operating lever by tightening them, and clamp the base plate tightly to the table with a screw. An error at this stage may cause the base plate to tilt during operation, which could lead to the thermometer breaking. Turn the friction band attached to the dynamometer around the friction cylinder 2.5 times (so that the dynamometer loosens when the lever is turned clockwise). Add a 5 kg weight to the lower end of the friction band.
2. To read the temperature, carefully insert the thermometer into the hole in the center of the friction cylinder. The hole in the cylinder and the thermometer must be carefully aligned; this is important to prevent the thermometer from breaking while the cylinder is rotating. To improve thermal contact, the hole in the cylinder is filled with a heat-conducting material.
3. At the start of the measurement, note the initial temperature reading and, by rotating the operating lever quickly and steadily 200 times, record the temperature in Table I every 30 seconds for 4 minutes. At the end of 4 minutes, when the temperature reaches its maximum, read the F_D force from the dynamometer. After the cycle is complete, record the temperatures in Table I every 30 seconds.
4. Draw the temperature graph as a function of time as shown in Figure 2. Read the temperature change value $\Delta T = T_2 - T_1$ from this graph.
5. 5. Using equations (3) and (5), calculate ΔQ . Using the F_D value read in step 3, calculate

ΔW . Find the ratio $\Delta W/\Delta Q$.

$$\frac{\Delta W}{\Delta Q} =$$

Table I:

t (s)	T (°C)	t (s)	T (°C)	t (s)	T (°C)	t (s)	T (°C)	t (s)	T (°C)
0		150		300		450		600	
30		180		330		480		630	
60		210		360		510		660	
90		240		390		540		690	
120		270		420		570		720	

Part II

Perform the experiment similarly and determine the heat capacity of the aluminum cylinder. Before starting the experiment, the friction cylinder and friction belt should be cleaned with a dry cloth to remove metallic wear dust. When using an aluminum cylinder, the friction force should not exceed 10 N to prevent heavy wear and staining. For this, use a dynamometer with a 1 kg weight and 10 N accuracy.

1. Repeat the steps in Part I of the experiment for the aluminum cylinder. Record the measurements in Table II.
2. Determine the approximate value of $\Delta T = T_2 - T_1$ from Table II.
3. Calculate the heat capacity (C) of the cylinder using Equation (8), with the number of rotations of the arm n, the temperature increase ΔT and the force F_D read from the dynamometer.
4. Calculate the specific heat capacity (c) of the cylinder using Equation (9).

Table I

t (s)	T (°C)	t (s)	T (°C)	t (s)	T (°C)	t (s)	T (°C)	t (s)	T (°C)
0		150		300		450		600	
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EXPERIMENT-RELATED QUESTIONS

1. What are heat and temperature? How can we distinguish between these two physical quantities?
2. . Debye's theory can accurately explain heat capacity at low temperatures. But is Debye's theory valid at high temperatures? Investigate.
3. Another theory proposed to explain heat capacity in solids is known as Einstein's theory. Recreate heat capacity using Einstein's theory. Draw the heat capacity expression you obtained as a function of temperature and compare it with the results of Debye's theory.
4. . What is latent heat? Is there a relationship between latent heat and heat capacity? Investigate.
5. In which of the reversible and irreversible processes is the heat change greater?
6. What are the laws of thermodynamics? Explain.
7. What physical consequences would you expect if a material had no heat capacity?
8. How can we increase the heat capacity of materials, mixtures, liquids, etc.? Have you ever thought about it?
9. What is a solar pool? Research it.
10. Can a material with increased heat capacity be used in industry, manufacturing, and for the benefit of society?
11. Interpret the graph you drew. Are the results as you expected?

REFERENCES

1. PHYWE series of publications, Laboratory Experiments